

Python Set Practice — Basic to Hard

50 Python Set Practice Questions

Arranged from Basic → Hard • Each question includes example input and expected output.

Level	Difficulty	Focus	Questions
1	Basic	Set creation, membership, size, add/remove, simple set operations	10
2	Basic → Intermediate	Union/intersection logic, filtering, comparisons, duplicates	10
3	Intermediate	Multi-step operations, subset/superset, symmetric difference, transformations	10
4	Advanced	Set-based problem solving, grouping, constraints, multi-step logic	8
5	Hard	Complex set reasoning and optimization-style problems	12

Level 1 — Basic

Set creation, membership, size, add/remove, simple set operations.

Q No.	Question	Example Input	Example Output
1	Create a set from a list of integers and remove duplicate values.	Input: [1, 2, 2, 3, 1, 4]	Output: {1, 2, 3, 4}
2	Find the number of elements in a set.	Input: S = {10, 20, 30, 40}	Output: 4
3	Check whether an element exists in a set.	Input: S = {2, 4, 6, 8}, x = 6	Output: True
4	Add an element to a set.	Input: S = {1, 2, 3}, x = 4	Output: {1, 2, 3, 4}
5	Remove an element from a set.	Input: S = {10, 20, 30}, x = 20	Output: {10, 30}
6	Find the union of two sets.	Input: A = {1, 2, 3}, B = {3, 4, 5}	Output: {1, 2, 3, 4, 5}
7	Find the intersection of two sets.	Input: A = {1, 2, 3}, B = {2, 3, 4}	Output: {2, 3}
8	Find elements present only in the first set.	Input: A = {1, 2, 3}, B = {2, 4}	Output: {1, 3}
9	Find elements present in exactly one of two sets.	Input: A = {1, 2, 3}, B = {3, 4, 5}	Output: {1, 2, 4, 5}
10	Clear all elements from a set.	Input: S = {1, 2, 3}	Output: set()

Level 2 — Basic → Intermediate

Union/intersection logic, filtering, comparisons, duplicates.

Q No.	Question	Example Input	Example Output
1	Check whether one set is a subset of another.	Input: A = {1, 2}, B = {1, 2, 3}	Output: True
2	Check whether one set is a superset of another.	Input: A = {1, 2, 3}, B = {1, 2}	Output: True
3	Find common elements among three sets.	Input: A={1,2,3}, B={2,3,4}, C={2,5}	Output: {2}
4	Find all unique vowels present in a string.	Input: S = 'programming'	Output: {'o', 'a', 'i'}
5	Find all unique words in a sentence.	Input: S = 'cat dog cat bird dog'	Output: {'cat', 'dog', 'bird'}
6	Count how many unique numbers are in a list.	Input: [1,1,2,3,3,3,4]	Output: 4
7	Find the missing number from 1 to n using a set.	Input: [1,2,3,5], n=5	Output: 4
8	Remove all duplicate values while preserving first-seen order.	Input: [4,2,4,1,2,3]	Output: [4,2,1,3]
9	Check whether two lists contain exactly the same unique values.	Input: A=[1,2,2,3], B=[3,1,2]	Output: True
10	Find values common to two lists using sets.	Input: A=[1,2,3,4], B=[3,4,5,6]	Output: [3,4]

Level 3 — Intermediate

Multi-step operations, subset/superset, symmetric difference, transformations.

Q No.	Question	Example Input	Example Output
1	Find the symmetric difference of three sets step by step.	Input: A={1,2,3}, B={2,3,4}, C={3,4,5}	Output: {1,2,5}
2	Find elements that occur in at least two of three sets.	Input: A={1,2,3}, B={2,3,4}, C={3,4,5}	Output: {2,3,4}
3	Find elements that occur in exactly one of three sets.	Input: A={1,2}, B={2,3}, C={3,4}	Output: {1,4}
4	Check whether two sets are disjoint.	Input: A={1,2}, B={3,4}	Output: True
5	Partition a list into unique even and odd values.	Input: [1,2,2,3,4,5,6]	Output: Even={2,4,6}, Odd={1,3,5}
6	Find duplicated values in a list using a set.	Input: [4,1,2,4,3,2,5]	Output: {2,4}
7	Find the first repeated value in a list using a set.	Input: [5,3,1,3,2]	Output: 3
8	Find numbers in A that are not in B and are greater than 10.	Input: A={5,12,15,20}, B={12,20}	Output: {15}
9	Find the number of distinct characters shared by two strings.	Input: S1='apple', S2='plane'	Output: 4
10	Find all words that occur in both sentences.	Input: S1='red blue green', S2='blue green yellow'	Output: {'blue','green'}

Level 4 — Advanced

Set-based problem solving, grouping, constraints, multi-step logic.

Q No.	Question	Example Input	Example Output
1	Find the longest consecutive sequence length using a set.	Input: [100,4,200,1,3,2]	Output: 4
2	Find all pairs of values whose sum is a target using a set.	Input: [2,7,11,15,3,6], target=9	Output: {(2,7),(3,6)}
3	Find missing and duplicate values when numbers should be 1..n.	Input: [1,2,2,4,5]	Output: Missing={3}, Duplicate={2}
4	Find common unique characters across three strings.	Input: 'abcd', 'bcde', 'cdef'	Output: {'c'}
5	Find the smallest set of values that covers all required categories.	Input: Values=['A','A','B','C','C'], Required={'A','B','C'}	Output: {'A','B','C'}
6	Group numbers into two sets with equal sums if possible.	Input: [1,5,5,11]	Output: {1,5,5} and {11} → Equal
7	Find all elements that appear in exactly k lists.	Input: Lists=[[1,2,3],[2,3,4],[2,5]], k=2	Output: {3}
8	Determine whether one list can be transformed into another by only removing duplicates.	Input: A=[1,2,2,3,3], B=[1,2,3]	Output: True

Level 5 — Hard

Complex set reasoning and optimization-style problems.

Q No.	Question	Example Input	Example Output
1	Find the maximum-length consecutive integer sequence without sorting.	Input: [9,1,4,7,3,2,6,8]	Output: 4
2	Find the intersection of many lists and return unique values.	Input: [[1,2,3],[2,3,4],[2,5,3]]	Output: {2,3}
3	Find all values whose frequency is exactly twice the frequency of another value.	Input: [1,1,2,3,3,4]	Output: Pairs={{(1,2),(3,4)}}
4	Solve the two-set difference after repeatedly applying a sequence of operations.	Input: A={1,2,3,4}, B={3,4,5}, Ops=['diff','union']	Output: {1,2,3,4,5}
5	Find the smallest number of distinct values needed to reach at least target sum.	Input: Values=[2,3,5,7,9], target=14	Output: 2 (5+9)
6	Find the union of all sets in a list of sets.	Input: [{1,2},{2,3},{3,4}]	Output: {1,2,3,4}
7	Find elements that appear in every set of a list.	Input: [{1,2,3},{2,3,4},{0,2,3}]	Output: {2,3}
8	Find the largest subset of numbers that are pairwise non-consecutive.	Input: [1,2,3,5,7]	Output: {1,3,5,7}
9	Check whether two sets have the same cardinality and no common elements.	Input: A={1,2}, B={3,4}	Output: True
10	Find the minimum number of set changes needed to make two sets equal.	Input: A={1,2,3}, B={2,3,4}	Output: 2
11	Find the union size of two large sets without constructing the union explicitly.	Input: A =20, B =15, A∩B =5	Output: 30
12	Find values that occur in exactly two of four sets.	Input: A={1,2}, B={2,3}, C={2,4}, D={3,4}	Output: {3,4}